

Global warming, glacial lakes and cloud burst events in Garhwal –Kumaon Himalaya: A hypothetical analysis

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ABSTRACT

Cloud Burst in Garwal-Kumaon Himalaya, in the Indian state of Uttarakhand, is a common phenomenon. Cloud Burst is a massive downpour associating with flashflood, landslide, and earth flow that causes enough devastation in mountain areas. However, in recent years the frequency of this event has been increased tremendously. One of the important causes of frequent cloud burst is faster evaporation rate from high altitude glacial lakes. The mean temperature of the Earth has been increased 0.75°C during last century due to global warming, resulting faster destruction of mountain or valley glacier. This phenomenon is already witnessed in Himalaya where glacier and glacial lakes are changing alarmingly. Glacier retreating rate in Himalaya is 10 to 60 m per year. A well number of small glacier ($<0.2\text{ sq km}$) have already disappeared. This glacier melts water and lake comes direct contact with clouds due to higher altitude, create favorable condition of cloud burst in high altitude areas of Uttarakhand. This paper is a Geographical approach to study the Cloud Burst events in Garwal-Kumaon Himalaya highlighting the role of glacier melt water and high altitude lakes on this catastrophic event.

Keywords: Cloud burst, Garhwal-Kumaon Himalaya, Global warming, Glacier melting, Glacial lakes.

1. Introduction

Cloud Burst is an excessive downpour with in short time over a smaller area. In plain areas water logged condition or localized flood can be occurred after cloud burst. But this situation can be worst in high altitude areas. Cloud burst is very much associated with flash flood, earth flow and landslides in high terrains. Cloud Burst is a normal event in Himalaya territory especially to Garhwal-Kumaon region of Uttarakhand. Almost every year various part of this region experience the cloud burst and associating disaster. But with the beginning of new millennium, the intensity and vulnerability of this event has been increased. Recently in 2013 cloud Burst was a one of the catastrophic event in human civilization. Though, the steep orography of Garhwal-Kumaun Himalaya and concentration of Monsoon clouds create the situation of cloud burst but these might not be the only factors. Now time has come to analysis it with emphasis on global climate change and its impact on Himalayan environment. Himalaya is the home of world's largest mountain and valley glaciers and source of several perennial rivers of Asia. There are also several hundreds of high altitudes lakes, 118 such lakes exist in Garhwal-Kumaon region only. Now, it is confirmed that mean temperature of the world's highest mountain increased 0.75°C to 1°C during last century (Bajracharya Samjwal Ratna et al 2007), so glacier and glacial lakes are changing alarmingly. These lakes are very sensitive to climate change and being increased in volume due to faster melting of glaciers. Due to strong solar radiation in high altitudes, evaporation from the lakes increase

the depth and volume of clouds which are collapsing in different pockets of Garhwal-Kumaon region as cloud burst.

This paper deals with geography of the Uttarakhand state, distribution of high altitude lakes in Garhwal-Kumaon Himalaya and their impact on cloud burst event. It is also deals with possible measures of the hazards associated with cloud burst. This paper is a hypothetical analysis based on secondary data, mainly IMD images, Map of High Altitude Lakes and data of present condition of Himalayan glaciers from ISRO used.

2. Geography of Garhwal –Kumaon Himalayan region

2.1 Location

The Garhwal and Kumaon Himalaya are located in northern state of India, Uttarakhand. The entire state lies between 28°44' to 31°28' N latitude to 77°35' to 81°01' E longitude. Total area of the state is 53,483 square Km constituting 1.63% of Indian landmass (http://www.nainitaltourism.com/Uttarakhand_Uttaranchal.html) and 93% of the state area is mountainous. This state has two divisions; name of the western part is Garhwal and Eastern part is known as Kumaon. The state capital is Dehradun, located in Garhwal division. These two divisions are not only separated by cultural heredity but also different with physiography and relief. Now, these two divisions of Himalaya are commonly known as Uttarakhand Himalaya. This entire mountain tract is located in between Tons River, tributary of Yamuna (west) and Kali River in East, boundary of Nepal. West to East length of this mountain region is near about 320 Km long and North-South width is near about 50Km, along the boundary of Tibet and UP.



Figure 1: Location Map of Uttarakhand

2.2 Physiographic features

Maximum portion (93%) of the Uttarakhand state is under hilly tract with rugged topography consists of snow clad mountain peaks; scatter crests, deep valleys, rapids, cascades cirques and glaciers etc. Though the entire state is under the Western Himalaya but there are two sub-divisions i.e. Garhwal and Kumaon Himalaya. Garhwal is on the west of Ganga river system and Kumaon is on left. Topography of Garwal division is comparatively rugged and

inhospitable than the Kumaon. The main mountain ranges are aligned in Northwest to Southeast direction (Kharkwal S.C 1971). Altitude varies from <1200m to >4800 m in this region.

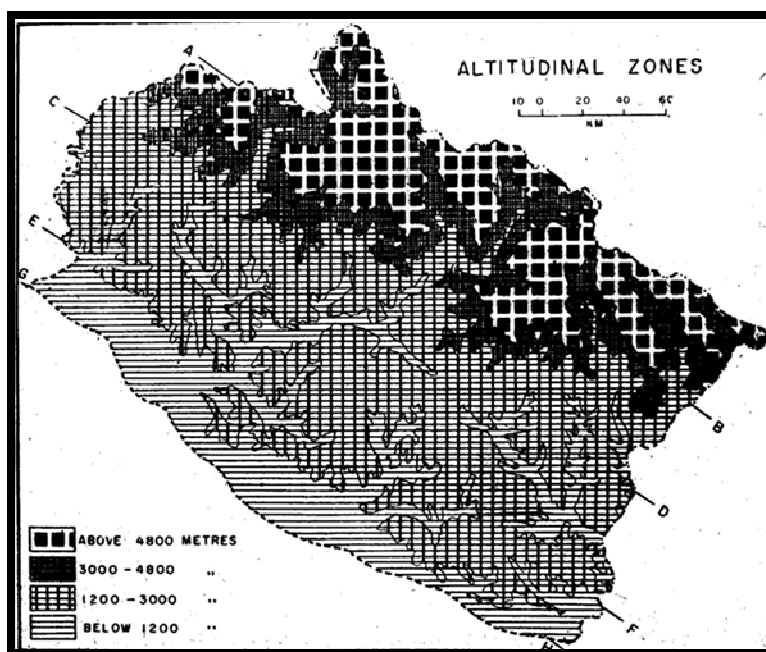


Figure 2: Altitudinal zones of Garhwal- Kumaon Himalaya
Source: India, A Regional Geography

The entire region is divided into three broad physiographic Regions;

Table 1: Physiographic Region of Garhwal & Kumaon Himalaya.

Major Physiographic Regions	Sub-Regions
Siwaliks (Sub Himalayan tract)	i) Siwalik ranges ii) Duns
Himachal (Lower Himalaya)	i) Himachal ranges and hills ii) Himachal valleys and Basins
Himadri (Greater Himalaya)	i) Himadri Ranges ii) Himachal Valleys

Siwalik Range is separated from the main range by the Main Boundary Thrust. Average altitude of this range is 750m to 1200 m. Southern slope is steeper and northern is gentle, stretched towards a structural valley, commonly known as Duns.

Himachal Range is a 75 km width massive mountainous tract, separated by Duns by the Main Boundary Thrust (MBT). The average altitude of this range is 1500m-2700m. Actually the whole range is a series of ridges, divided from each other by deep valleys of rivers. There are several lakes in Kumaon area, mainly structural basin. These lakes are commonly known as Tal.

Highest range of Himalaya is Himadri width 50 km width and average height is varies from 4800m to 6000m. There are four prominent groups of mountain ranges. These are; a) Bandar Punch (6315 m); b) Gangotri (6614 m), Kedarnath (6940m), Chaukhamba (7138); c) Kamet (7756 m); d) Dunagiri (7066 m), Trishul (712 m), Nnda Kot (6861 m). (Kharkwal S.C

1971). These four groups of peaks are separated with each other by deep valleys of Bhagirathi, Alokendra, and the Dhaul Ganga. There are 118 in high altitude glacial lake in Himadri ranges (National Wet Land Atlas: High Altitude Lakes in India, 2012) which have great environmental and ecological importance.

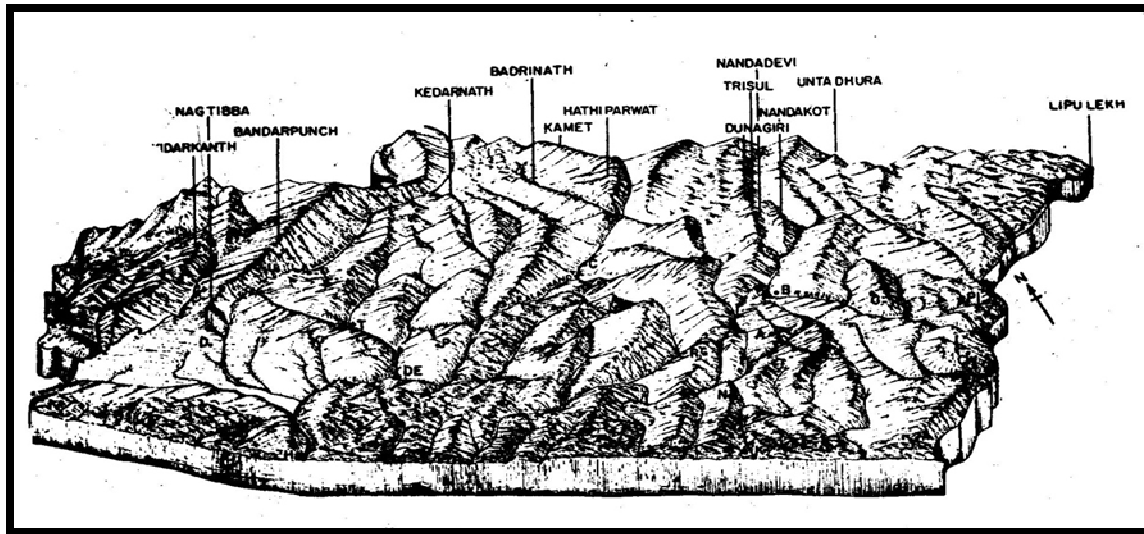


Figure 2: Block Diagram of Garhwal-Kumaon Himalaya.
Source: India- A Regional Geography

2.3 Climate

Climatic condition is markedly different in high altitudinal areas and the lower basins in Uttarakhand. Temperature is varied not in different seasons but also with the altitude. For example, in summer the valleys experience hot steamy tropical climate, while at a distance of about 75 km, the great range bears one of the highest snow-field of the world. Normally January is the coldest month (Kharkwal S.C 1971). Snowfall and chilly cold is a common phenomenon in high altitudes and lower plane experience mild temperature during winter. Snow fall for seven to eight days in each three months January to March is a regular event due to western disturbance. July is the hottest month of northern part of the state with average temperature near about 7°C, where as May is the hottest month in southern part with average temperature 35°C. According to altitude the entire state is divided into 7 micro climatic regions, these are;

Table 2: Micro climatic zones of Uttarakhand

Climatic zones	Altitude in meter	Temperature in °C		
		Annual	June	January
Tropical Zone	300-900	21.1	29.4	13.3
Warm Temperate Zone	901-1800	18.9	27.2	11.1
Cool Temperate zone	1801-2400	13.9	21.1	6.1
Cold Zone	2401-3000	1.3	17.2	2.8
Alpine zone	3001-4000	4.5	13.3	Below zero for 6 months, snow melts in July & August
Glacial zone	4001-4800	Ten months below zero two month between 2.2-3.9		
Perpetually frozen zone	Above 4800	Cold desert no vegetation		

Source: India- A Regional Geography

The Monsoon arrives in end of the June and retreats normally in middle of September. Nowadays, cloud burst during the Monsoon and pre-Monsoon is a regular matter in high altitudinal area of Garhwal and Kumaon Himalaya. Decreasing trend of annual rainfall (-29.7—2.1 cm/100 years) has been observed at Srinagar, Mussoorie, Mukteshwar and Joshimath whereas increasing trend at Dehradun, Pauri, Nainital, Almora and Pithoragrah (3.8 to 28.7 cm/100 year) has been observed in last century (Borgaonakar et al.1998). Mean temperature of this region has been increased 0.75° C to 1 °C during the last century due to global warming. (Bajracharya Samjwal Ratna et al 2007).

3. Record of cloud burst events in recent past in Garhwal &Kumaon Himalaya

Cloud burst is an extreme amount of precipitation over a smaller area within a shorter period. Sometime hail and thunderstorm are associated with it and can create flood or flash flood conditions causing loss of life and properties. It is observed that during the Cloud Burst the rate of rainfall can be reached up to 10 cm/hr (Woolley Ralfer, 1946). It is a localized phenomenon affecting an area not more than 20-30 square km but some time it may be 50 square km. Like the other meso scale weather structures such as tempest, tornado cloud burst has also shorter life span unlike the tropical cyclone and very difficult to predict. Cloud Burst is not an uncommon event in Uttarakhand Himalaya. In recent past this region has experienced several Cloud Burst incidences such as;

Table 3: Record of cloud burst in Uttarakhand, Garhwal-Kumaon Himalaya.

Affected Area	Year of Occurrence	Rate of Destruction
Kedarnath, Rudraprayag District	14-15 th June,2013	10,000 were died and 7000 missing and several injured
Rudraprayag, Rudraprayag District	14 th September,2012	39 were people died.
Munisyari, Pithoragarh district	18 th August,2010	30 were died and 50 injured
Kapkot village,Bageshwar District	19 th August,2010	18-20 were died, more than 30 missing
Nachni near Pithoragarh District	7 th August,2009	38 were people died.
Near Badrinath Shrine, Chamoli District	6 th July,2006	17 people died and 28 injured
Malpa, Kali Valley of Kumaun Division	17 th August,1998	250 were died including 60 Kailash Mansarovar pilgrims
Badrinath Shrine, Chamoli District	24 th July,2004	17 were died and 18 missing
Alakananda river basin	July,1970	15,000 were died and 80,000 houses were destroyed along the bank of Alakanada.

Source: Wikipedia

4. High altitude lakes and cloud burst event in Garhwal-Kumaon Himalaya

Garhwal-Kumaon Himalaya is a suitable place for cloud burst. Cloud burst is mainly occurred here in Monsoon. When Monsoon clouds are obstructed in main Himalayan range, the clouds rose upwards (sometime up to 9 km), form dense Cumulonimbus cloud with wider base. It is happened due to moist thermodynamic instability and rapid dynamic lifting of cloud by step topography. Cloudbursts can't occur only in the Monsoon period but also may

occurred during the March to May as pre Monsoon Showers. Nowadays, intensity and destruction of cloudburst are being increased in Garhwal-Kumaon region. One of the major factors of the frequent cloud burst is faster melting of glacier and expansion of Glacier lakes. There are 118 high altitudinal glacial lakes in Uttarakhand, occupying 231ha, less than 1% of total wetland of Uttarakhand (National Wet Land Atlas: High Altitude Lakes in India). These 118 high altitudinal lakes are located in only 4 districts (Chamoli, Pithoragarh, Rudraprayag and Uttarakashi) among the 13 districts of Uttarakhand. Maximum numbers (60) of high altitude lakes are located in Chamoli districts followed by Uttarakashi 32. Only one lake is present in Rudraprayag district. Most of the lakes are very small in size (<2.5 ha). Some famous lakes of these Himalayan regions are Kedar Tal (4425 m), Vasuki Tal (4135 m), Chorabari Lake (4220 m), Shasra Lake (4572 m), Rupkund (529 m), Hemkund (4572 m) etc. Among the 118 lakes of this region 33.9% are present within 3000-4000 m, 57.62% situated within 4000 to 5000 m and 8.47% exist above 5000 m altitude (National Wet Land Atlas: High Altitude Lakes in India). Though, the sizes of the high altitude lakes are small but have great environmental impacts. Size and altitude wise distribution of these 118 glacial lakes are given in table no 4;

Table 4: Altitudinal range-wise distributions of high altitudinal lakes of Garhwal and Kumaon Himalaya

Sl. no	District	Altitudinal Variation						Total	
		(3000-4000m)		(4000-5000m)		(>5000m)			
		No. of Lakes	Area in ha	No. of Lakes	Area in ha	No. of Lakes	Area in ha	No. of Lakes	Area in ha
1	Chamoli	26	28	26	63	8	21	60	112
2	Pithoragarh	3	3	20	65	2	8	25	76
3	Rudrapr rayag	1	2	-	-	-	-	1	2
4	Uttarakashi	10	10	22	31	-	2	32	41
	Total	40	43	68	159	10	29	118	231

(Source: National Wet Land Atlas: High Altitude Lakes in India, ISRO)

Table 5: Size-wise distributions of high altitude lakes of Garwal-Kumaon Himalaya

Districts	Size of the Lakes												Total	
	>500ha		100-500ha		25-100ha		10 -25ha		<10 ha		<2.5 ha			
	No. of Lakes	Area in ha	No. of Lakes	Area in ha	No. of Lakes	Area in ha	No. of Lakes	Area in ha	No. of Lakes	Area in ha	No. of Lakes	Area in ha	No. of Lakes	Area in ha
Chamoli	-	-	-	-	-	-	1	17	13	49	46	46	60	112
Pithoragarh	-	-	-	-	-	-	-	-	11	62	14	14	25	76
Rudraprrayag	-	-	-	-	-	-	-	-	1	2			1	2
Uttarakashi	-	-	-	-	-	-	-	-	3	12	29	29	32	41
Total	-	-	-	-	-	-	1	17	28	125	89	89	8	231

(Source: National Wet Land Atlas: High Altitude Lakes in India, ISRO)

Maximum numbers of high altitude lakes are fault basin, evolved due to orographic movement in Holocene period. These lakes are fed by glacier melt water as maximum are located in snout areas and also receive enough precipitation by Monsoon. The mean annual

precipitation ranges from 600-1200 mm and 80% of it is occurred in Monsoon (Climate Himalaya 30th July 2013). The high altitude lakes are very sensitive to climate change. These lakes are characterized by extreme cold, dry and alpine climatic conditions and covered with seasonal or diurnal permafrost layer. This high altitude area experience low air pressure and higher ultraviolet radiation in cloud free conditions. Due to strong solar radiation the volume of the lakes are expanded with glacier melt water in summer. During winter these lakes are frozen with snowfall due to western disturbance with presence of extra tropical depression. The freezing period generally extended from September to April. In such a way the freezing and melting cycle is generating with mixing of vertical water column.

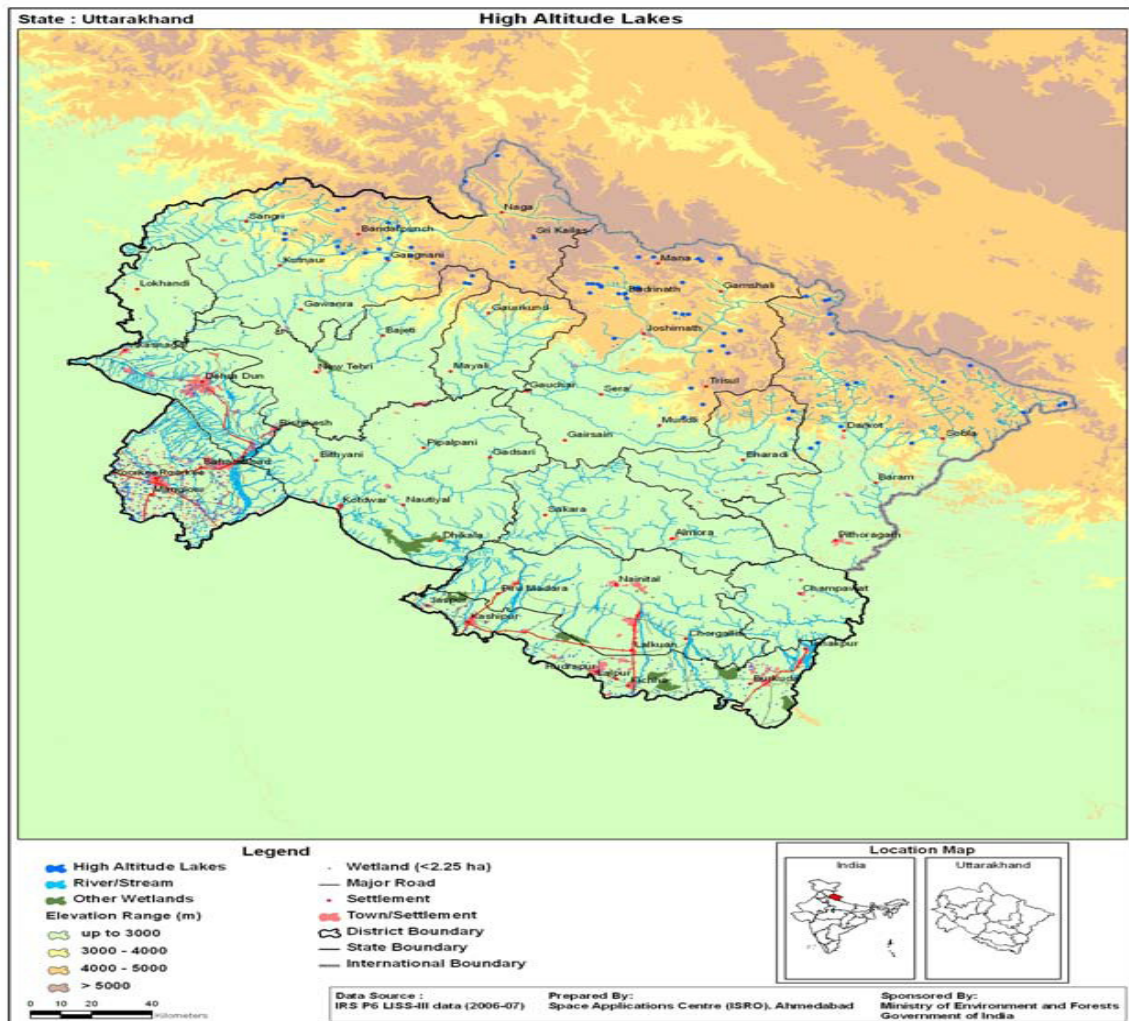


Figure 3: Distributions of High Altitude Lakes of Uttarakhand
(Source: National Wet Land Atlas: High Altitude Lakes in India, ISRO)

The mean temperature of the planet has been increased 0.75° C during last century and would be increased 1.4 to 5.8° C due to global warming, resulting faster destruction of mountain or valley glaciers. (Ray Mukesh et al, 2011) This phenomenon is already witnessed in Himalaya where glacier and glacial lakes are changing alarmingly. Glacier retreating rate of Himalaya is 10 to 60 m per year. A well number of small glacier (<0.2 sq km) have already disappeared. (Ray Mukesh et al, 2011) Some glacial lakes in this region have been increased as high as 800 percent over the last 40years.(Ray Mukesh et al, 2011) At present a series of sub-glacial ponds are evolving and merging with each other so that glacier are melting faster than the past.

Table 6: Present status of Glaciers of Garhwal –Kumaon Himalaya

River Basin	Coverage	Trend in Glacier Area Estimation	Status of Studies Glacier
Alakananda (Satopanth & Bhagirath Karak are 2 large glacier of this basin)	274 glaciers studied	14% loss in area (During 1962 to 2001 the area of glaciers was 1047 & 905 Square km respectively)	243 showed retreat (88.68%), 4 showed no change, 27 glaciers advanced. Glacier retreat for this basin after 1990 is much rapid.
Bhagirathi	183 glaciers studied	11% loss in area (During 1962 to 2001 the area of glaciers was 1218 & 1074 Square km respectively)	117 showed retreat (64%) 39 showed no change 27 glacier advanced Glacier retreat has been slow after 1990 Adjacent but contrasting trends
Gauri Ganga (Milam Glacier)	29 glaciers studied	4% loss in area (During 1962 to 2001 the area of glaciers was 272 & 261 Square km respectively)	Most glaciers of this basin slow retreat
Dhauliganga	104 glaciers studied	16% loss in area	

Source: Snow and Glaciers Studies of the Himalayas, Space Application Centre, Ahmedabad (May 2011); Monitoring done through topographic maps of Survey of India by SAC.



Kedartal in summer



Kedartal in winter

This glacier melts water and lake water come direct contact with clouds due to higher altitude. The temperatures of the glacial lakes are normally below the freezing temperature. The saturation vapour pressure over cold ($>-5^{\circ}\text{C}$) water of lakes remains high (Figure 4). So, the evaporated water of the lake immediately condenses over the snow droplets of the cloud. Moreover, the volume and depth of the cloud could be increased within short time by sublimation process from the freeze lakes or moraine free glaciers in snout areas. Garhwal-Kumaon region is highly comprises with micaceous Pinjor sediments (Chaudhri A.R, Sing Mahavir 2012) such as Gneiss, Mica Schist and Garnet Schist, Biotite etc. So, there are plenty numbers of freezing nuclei (mainly Illite, keolinite) presence in atmosphere due to excessive weathering and erosion process of these micaceous elements. This freezing nuclei rapidly freezes small water droplet into ice crystal within 0°C to -40°C temperature and

cloud become more dense and heavy. In such a way large wide based cumulonimbus cloud formed. When this heavy cloud moves downwards, become instable due to adiabatic heating and can be collapsed (as a balloon full of water) with in small portion of area. When this type of cloud is trapped in a valley, the rate of destruction is become worst. Moreover, the situation can be more critical with the presence of extra tropical depression like 15th June 2013 (Das Pranab Kr. 2013).

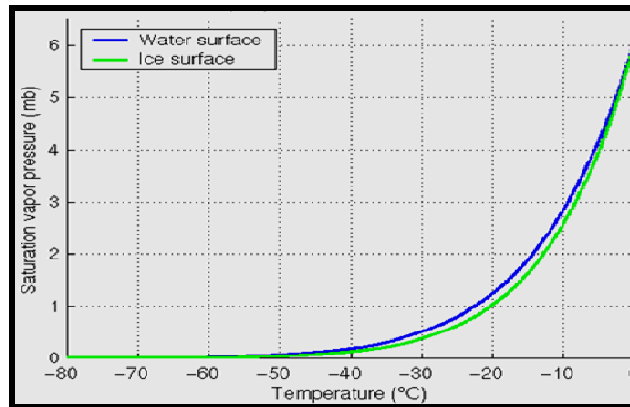


Figure 4: Saturation vapour pressure over water and ice droplets

Source: http://www.geminibv.nl/tidbits/dauwpunt-druk-en-temperatuur?set_language=en

5. Cause of cloud burst disaster in 2013

In 2013 disaster there was an active extra tropical depression (arrived from Black sea-Caspian Sea region) over the Uttarakhand and adjoining areas, just before the arrival of South Western Monsoon. In 2013, Monsoon arrived before the normal time. This extra tropical depression pulled the Monsoon towards western Himalaya. Due to collusion between two air-masses (Figure 5), the formation of cumulonimbus cloud generated rapidly. More over the faster rate of evaporation of glacial lakes in Uttarakhand and surrounding Himachal Pradesh and Nepal increased the volume of the cloud. The concentrations of the clouds were so high that there was a multiday cloud burst in Kedar Dome and surrounding areas. The WIGH (Wadia Institute of Himalayan Geology) meteorological observatory at Chorabari Glacier camp (3820 m) recorded 325 mm rainfall on and from 15th June to 16th June and surface air pressure reached 832.4 mb in 17th June. (Climate Himalaya 30th July 2013)

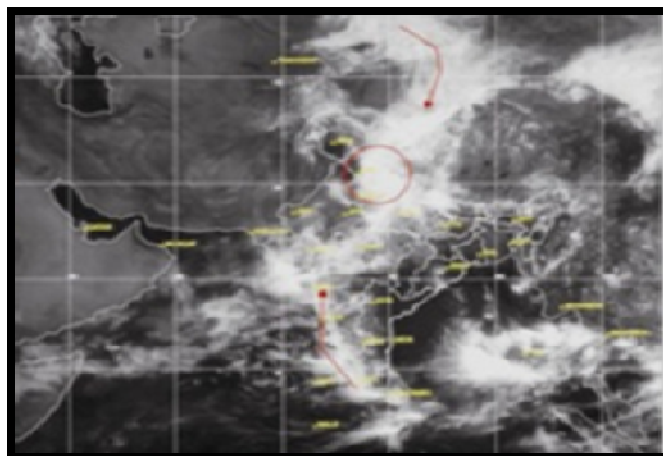


Figure 5: IMD image (17.06.2013) shows collusion of two different air masses over Garhwal –Kumaon Himalaya

Kedarnath and surrounding areas are under the Pindari Thrust comprises with gneiss, schist and biotite, extremely affected by physical weathering of kindreded shattering; debris slide-slump and earth flow etc. But most important is fluvio-glacial erosion. Kedarnath is located on the paleo-channel or in a narrow immature flood plain in between Mandakini and Saraswati Rivers; both are originated from Chorabari Lake. During the multiday cloud burst (14th June to 17th June) million tons of debris, boulders have been carried down with the collapsed water of Chorabari Lake through Mandakini-Saraswati Rivers as flash-flood. Continues downpour also increased the rate of landslides in fragile mountains. (Das Pranab Kr. 2013).

5.1 Conclusion and suggestion

2013 Cloud Burst event was a great lesion. It is very important to predict cloud burst with proper early warning system. If it would be possible the casualty of life could be minimize. We have to also normalize the environmental quality in high altitude areas so that mountain glaciers exist with their beneficial effects. It is true that durability of Glaciers in Grhwal-Kumaon Himalaya is in threat due to global warming. However, localize factors such as excessive practice of mass tourism, vehicular movement, human activities in high altitude eco-sensitive areas also responsible for increasing of temperature and destruction of glaciers. With the implementation of some techniques, rules and regulation intensity and vulnerability of cloud burst and associated hazards could be minimized

1. It is difficult to forecast a mesoscale weather phenomenon especially cloudburst. But to some extent, now it is possible. But it requires high-resolution numerical models and mesoscale observation equipments such as of high performance computer, Doppler Weather Rader (DWR), Automatic Weather Station (AWS), Radiosonde / Rawinsond (RS/RW) etc (Das Someshwar et al, 2006). These observation equipments can be installed in every sub divisional towns for high resolution measurement (>10 km) of vertical profiles and parameters of atmosphere. This entire model works in NOWCAST mode only i.e. few hours in advance. So, education and tanning of the local administration should be incorporated for preparation within short notice that helps mitigation of the disaster.
2. For continues monitoring the glacier, more observatory centers should be establish in snout areas with Automatic Weather Station (AWS) and data from the AWS should be generated and studied in a mother station with a regular interval. In this context, WIGH is doing excellent job. However, number of observatory centers should be increased immediately.
3. With the early warning system, effective evacuation plans and responsive disaster management group should be prepared with the proper guidance and assistance with National Disaster Management Authority (NDMA). These groups can be deployed block wise with active Participation of local inhabitants (Das Pranab Kr. 2013).
4. There should be strict restriction on mass tourism in Eco-sensitive zones especially in Char Dhams, Valley of Flowers, Hemkund Sahib and several glaciers and snouts areas. Annual quota of tourists or pilgrims should be introduced in these areas like Kailash –Manasarovar Yatra.
5. Considerable amount of environmental tax must be imposed on tourists and vehicles to regulate the influx vehicular movement in high terrain areas.

6. The state authority can imposed restriction in night halt in eco sensitive zones. If it will be possible, the emulsion of green house gases and melting rate of the glacier in snout areas could be minimized. Spreading of pollutants such as plastics, other solid waste and chemicals could be less with this restriction in catchment area of the river basins.
7. Awareness programmes must be enhanced within the tourists and pilgrims to normalize the environment of high terrine areas. Well trained local people can play an important role in this programme.
8. Heavy Duty vehicles should be prohibited in slide porn areas and regular monitoring should go on with the probation of fine. Annual quota could be introduced for registration of all types of vehicles to regulate the new vehicle. All the vehicles that are older than 10 years should be ban for better air quality throughout the state.
9. It is observed that the transport syndicate or agencies of UP and Delhi, surrounding states, operate the vehicle for Char Dham Yatra and mass tourism and they illegally operate more vehicles than their authorized permission. The transport syndicates or agencies from outside of Uttarakhand should be banned or restricted with probation of state wise quota. The illegal transport syndicate or agencies should be ban permanently in case of violation of terms and conditions.

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